

EEG scalp data processing using FIR filter, Kaiser windowing technique with features extraction and classification for epileptic seizure detection

Aruna Pant, Adesh Kumar ^{*} 

Department of Electrical & Electronics Engineering, School of Advanced Engineering, University of Petroleum and Energy Studies (UPES), Dehradun, Uttarakhand, 248007, India

ARTICLE INFO

Keywords:

Kaiser window
FIR filtering
Notch filter
Epileptic
EEG signal processing

ABSTRACT

Epileptic seizures are caused by abnormal synchronized discharges of neuronal activity in the brain, which can be recorded from the scalp using multichannel EEG signals. Traditionally, neurologists interpret these signals manually, a process that is both time-consuming and dependent on their availability. This study proposes a fully automated method to distinguish seizure from non-seizure EEG signals using advanced signal processing and machine learning. The EEG signals are first filtered using a Kaiser window-based notch filter, which is particularly effective at reducing spectral leakage and removing line frequency noise (50–60 Hz). The performance of different Kaiser window β values (4, 6, 8, and 12) is analyzed, with $\beta = 12$ showing the best results due to minimal leakage, high side-lobe attenuation, and narrow main lobe width, indicating high filter selectivity. The filtered EEG signals are characterized using six statistical features: mean, mode, median, variance, kurtosis, and skewness. These features are then used to train a machine-learning classifier to detect epileptic activity automatically. The novelty of this work lies in the optimized filtering using the Kaiser window and its integration with statistical feature-based classification, providing an efficient, automated seizure detection system that minimises manual intervention.

1. Introduction

The Electroencephalography (EEG) measures irregularities in the brain's electrical activity. Electrodes are implanted on the scalp to track and record brain wave patterns. The results are then recorded on a computer [1]. Abnormal EEG patterns indicate seizures and other issues. Non-invasive electrodes for EEG are usually applied to the scalp with a cap or individual electrodes (e.g., the 10-20 system), whereas invasive electrodes are implanted directly into the brain, frequently for patients with epilepsy or other problems [2]. EEG signals with small amplitudes of 10 to 100 (μ V) might be polluted by artifacts and lose their uniqueness during recording. The most prevalent disturbance is power-line frequency noise at 50 Hz [3]. This makes clinical analysis and data retrieval challenging. To make an accurate diagnosis, all such abnormalities in EEG readings must be removed. Artifacts are undesired electrical signals that come from non-cerebral sources and can be divided into two categories: extraphysiological artifacts, which come from outside sources like equipment interference, environmental electrical noise, or electrode malfunctions, and physiological artifacts, which come from the patient's

own body, such as eye movements, muscle activity, or cardiac rhythm. The term 'Disturbances' describes outside interferences that impact the EEG signal but are not related to the recording equipment or the physiology of the individual. Filtering is a key feature of digital signal processing [4]. In signal processing, there are many circumstances in which the input signal to a system has extra unneeded noise, degrading the quality of the original signal [5]. The FIR filter is a fundamental component of digital signal processing systems [6]. It ensures a linear phase frequency characteristic with frequency characteristics. The unit impulse response of an FIR filter is of finite duration. This study utilized various FIR window approaches to separate the EEG signal into its main components (Delta wave, Theta wave, Alpha wave, and Beta wave) [7]. When a finite portion of a signal is processed (for example, filtering), it is assumed that the signal repeats infinitely outside of this segment. This repetition causes discontinuities along the segment's edges, resulting in spectral leakage in the frequency domain. Window functions are used to smooth out the ends of a signal segment, minimizing discontinuities and consequently spectral leakage. Distinct window functions [8] have distinct qualities, and the choice is based on the application's specific

This article is part of a special issue entitled: Wearable Innovations for Human Augmentation published in Measurement: Sensors.

^{*} Corresponding author.

E-mail addresses: adeshmanav@gmail.com, adeshkumar@ddn.upes.ac.in (A. Kumar).

<https://doi.org/10.1016/j.measen.2026.101998>

Received 25 June 2024; Received in revised form 27 February 2026; Accepted 28 February 2026

Available online 2 March 2026

2665-9174/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

requirements. Kaiser window is a flexible window function that strikes a balance between main lobe width and side lobe suppression. The trade-off between side lobe attenuation and main lobe width in the frequency domain is controlled by the β (beta) parameter in the Kaiser window. A non-negative real value, β , modifies the tapering of the window. It controls the rate at which the window tapers away from the center. The greater value of β leads to improved frequency selectivity due to a narrower main lobe and reduced spectral leakage due to increased side lobe attenuation [9].

The frequency response of FIR filters in DSP is largely shaped by window functions. Filter performance is directly affected by the unique properties of each window function, including stopband attenuation, side lobe, and main lobe width. These characteristics affect the filter's capacity to reduce spectral leakage and resolve closely spaced frequency components. Applications needing efficient noise suppression are better suited for the Hamming window since it provides better stopband attenuation than the rectangular window. On the other hand, where shorter filter lengths or less computational complexity are important, windows such as Bartlett or rectangular are often preferred. Therefore, the selection of a window function largely depends on the specific requirements of the application, particularly the trade-offs between frequency resolution, computational efficiency, and artifact suppression, especially relevant in EEG signal processing [10]. The Kaiser window stands out for its flexibility, offering a tunable parameter (β) that allows precise control over the balance between main lobe width and side lobe attenuation. This makes it especially suitable for tasks like EEG signal denoising, where both adaptability and high performance are essential.

The study contributes to the fully automated technique for differentiating between seizure (epileptic) and non-seizure EEG signals [11]. This expedites diagnosis and lessens reliance on neurologists, particularly for lengthy recordings. The study systematically assesses how well the Kaiser window works with various β values (4, 6, 8, 12) while creating a notch filter to eliminate powerline noise at frequencies between 50 and 60 Hz. The analysis considers the factor of leakage, attenuation of the relative side lobe, and width of the main lobe (mHz). It enhances the precision and effectiveness of epileptic signal detection and provides a new comparative understanding by providing a methodical assessment of Kaiser window parameters for EEG filtering. It creates a whole end-to-end pipeline from raw EEG to decision-making, bridging the gap between signal processing and machine learning categorization.

The organization of the manuscript is as follows: The relevant review is given in section 2, followed by the methodology in section 3. The proposed approach focuses on filtering EEG signals and detecting epileptic seizures with improved accuracy. Section 4 consolidates the outcomes of filtering, feature extraction, and classification with statistical analysis to support the effectiveness of the proposed method for epileptic seizure detection. The key findings are concluded in section 5, including the efficacy of the FIR Kaiser window-based notch filter and ANN classifier in enhancing EEG signal quality and seizure detection accuracy.

2. Related work

Digital filters are classified into two types: Infinite Impulse Response (IIR) and Finite Impulse Response (FIR) filters. In the FIR filter, the impulse response has a limited time, whereas in the IIR filter, it has an infinite duration. The FIR filter has a symmetrical tap of length (N) and produces a linear phase. It is simple to design, stable, and excellent for phase-sensitive applications [12]. Numerous studies have compared the efficiency of window-based FIR filters in attenuating artifacts from biological signals. A comparative analysis of six window functions for FIR notch filtering was presented in Ref. [13], highlighting the Kaiser and Blackman windows. In Ref. [14], the Harris window was shown to outperform the Gaussian and Blackman windows in suppressing power-line interference in ECG signals. Various windowing approaches,

including Kaiser, Blackman, Hanning, and Hamming, have been considered for building FIR filters. The suggested Kaiser window method efficiently removes noise from EEG signals due to its short length, maximal stopband attenuation, and minimal transition width, compared to other methods [15]. The 100th-order digital FIR and IIR filters were applied to signal epochs, and performance was analyzed using the fast Fourier transform and signal-to-noise ratio [16]. Finite impulse response filters outperform IIR filters in removing noise from EEG signals. The objective of this research is to eliminate random noise in EEG data [17]. FIR digital filters with a Kaiser Window to remove the interference or the artifacts from the ECG signal [18]. The use of an impulse response filter to substitute noise generated before and after the FIR filter is applied [19]. The 50 Hz powerline noise is reduced with a band-pass filter with a frequency range of 0-40 Hz. Power spectral density (PSD), kurtosis, and skewness were utilized to assess the EEG signal after filtering [20]. Conversely, the notch filter eliminates activity within a very narrow range. The typical oscillation frequency of the alternating current in standard electric outlets is 60 Hz. The EEG is typically contaminated by the electromagnetic fields so created. 60 Hz notch filters are used to address this specific problem [21]. The FIR filter removes powerline interference from the noisy EEG signal and baseline drift. It makes use of a 64-tap FIR, which has a high dynamic range and a 16-bit coefficient [22]. A low-pass FIR filter reduces noise that frequently overlays an EEG signal and increases SNR [23]. To remove artifacts from EEG signals, conventional filters such as Chebyshev, Butterworth, FIR bandpass filters, and adaptive recursive bandpass filters are also used [24]. Two FIR and IIR are features of digital filters. FIR filters are stable, have a symmetrical tap of length (N), offer a linear phase, are easy to construct, and work well in applications where phase is critical [25]. The noise in the EEG signal is also removed using a digital filter [26]. In Ref. [27] contributed to EEG data processing by comparing various windowing methods in the design of FIR filters, improving signal filtering techniques for better analysis, and [28] contributed by exploring soft computing approaches for processing human scalp EEG, advancing methods for efficient signal classification and analysis. A suitable window function method can be used to lessen Gibbs oscillations. Many windowing strategies have been presented by researchers to filter EEG data; however, only a few provide good solutions for removing noise from EEG data [29,30].

This research introduced a computer-assisted EEG analysis technique for the diagnosis of epilepsy [31]. They employed SVM for classification after feature extraction in the time-frequency domain. The model was not verified on extensive or varied datasets and showed a lack of resistance to noise distortions. The paper [32] presented a methodology that integrates wavelet transformations and machine learning to classify EEG signals. Wavelet-based techniques employed SVM and KNN. Performance deteriorated with high-dimensional data and failed to account for real-time implementation factors. An automated system for the detection of epileptic seizures was created utilizing feature selection and classification techniques [33]. They employed correlation-based feature selection in conjunction with Random Forest classifiers. The model's generalizability to individual patient variances was not explored. The comparative analysis assessed various classification methods employing wavelet packet decomposition [34]. They evaluated SVM, KNN, and decision trees for EEG categorization. Discrepancies are identified the practicality of real-time application and adaptability to unfamiliar issues was not investigated. An automated EEG classification system utilizing iterative filtering was created, incorporating filter-based preprocessing and machine learning classification, although the specific classifier was not the primary focus [35]. The study was deficient in cross-validation among patient datasets and failed to consider computing efficiency. Automatic EEG channel selection emphasized supervised learning for feature selection in conjunction with SVM and other classifiers [36]. The method's responsiveness to fluctuating EEG readings during seizures requires enhancement. An EEG categorization system for seizure prediction was suggested in Ref. [37]. A hybrid model of CNN and LSTM

was employed for deep learning analysis. The system was deficient in energy-efficient implementation and had not undergone testing for real-time performance. Random Forest model optimized using grid search for the detection of epileptic activity [38]. The strategy failed to account for patient-specific adjustments or computational lag in embedded systems. A hybrid categorization strategy was devised for individual patients [39]. They derived from a synthesis of support vector machines and decision trees with enhanced electrode selection. The complexity of the hybrid model may impede its real-time clinical application, and scalability was not evaluated. Concentrated on the hardware-efficient realization of a neural classifier for closed-loop neuromodulation [40] and employed a Neural Tree Classifier optimized for on-chip implementation. Despite its potential for hardware efficiency, the adaptability to various EEG signal patterns remains untested. Despite extensive research on FIR and IIR filters for EEG signal denoising, there is a lack of real-time, adaptive filtering techniques optimized for varying EEG artifacts and powerline noise. Additionally, many studies overlook, filter generalizability, and performance evaluation beyond basic signal metrics.

The reviewed studies collectively demonstrate significant progress in EEG- and neuroimaging-based intelligent healthcare and brain-computer interface (BCI) systems, spanning signal generation, preprocessing, feature learning, and clinical applications. Early work focused on modelling and embedded implementations, such as artificial EEG generation that emulates neurological artifacts and suppresses harmonics [41], and adaptive, FPGA-based BCI architectures using dynamic filtering and classical CSP-LDA pipelines for motor imagery tasks [42].

Subsequent research emphasized subject-specific and self-adaptive classification strategies to address inter-subject variability and improve robustness in online BCI operation [43], along with multi-class EEG frameworks enabling fine-grained motor control, such as individual finger movement detection for prosthetic applications [44]. More recent studies increasingly adopt deep and hybrid learning paradigms, integrating handcrafted, temporal, spectral, and deep features with CNNs, BiLSTMs, attention mechanisms, and interpretable ensemble models to enhance performance and clinical trustworthiness in epilepsy, emotion recognition, and neurological disorder diagnosis [45–47]. The study in Ref. [48] highlights that the choice of window-based filtering methods plays a crucial role in improving EEG signal quality by effectively managing noise and enhancing signal reliability. The work in Ref. [49] shows that carefully designed EEG preprocessing and filtering steps substantially improve the clarity of epileptic-seizure-related patterns in the signal. The study further reports that these enhanced features lead to more reliable and accurate classification performance in automated seizure-detection systems [50]. These works collectively show the progression from embedded BCI feasibility to advanced EEG analytics. Real-time device control was demonstrated in an embedded setup [51], while adaptive filtering improved motor-imagery classification [52]. Low-latency SMR detection was achieved via zero-time windowing [47], and brain-inspired features enabled reliable stress recognition [53].

3. Methodology

The following methodology is used for removing the line

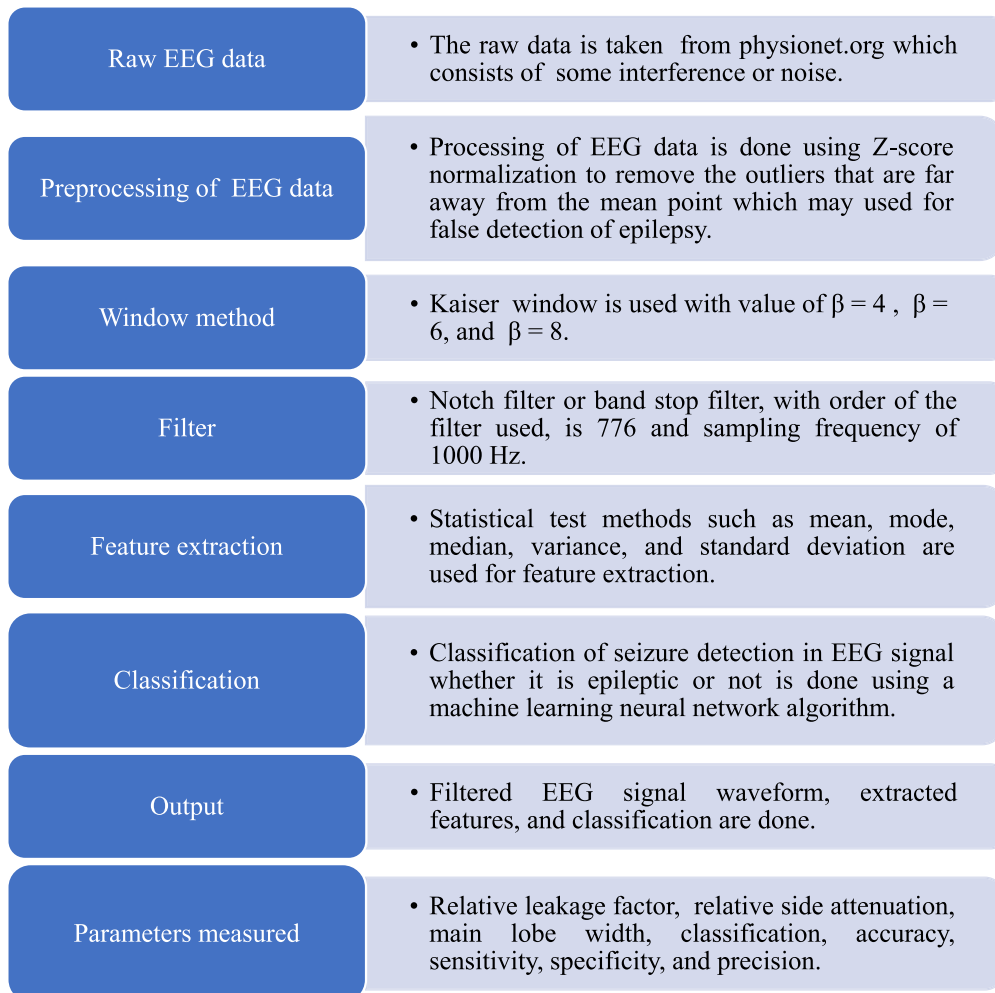


Fig. 1. Methodology for filtering EEG signals.

interference noise of 50-60 Hz from Scalp data. First, the raw EEG data is loaded from [Physionet.org](https://physionet.org), and the steps given in Fig. 1 are followed. As the EEG data is nonstationary and has a very small amplitude in the microvolt range, it is easily affected by noise. The methodology diagram essentially outlines the pipeline for building an EEG-based seizure detection system, from raw data collection to classification and performance evaluation. The first step is to process the raw EEG data. The EEG data is sourced from physionet.org, a public database. The raw signals usually contain some noise or interference that needs to be addressed. Step 2 is for the preprocessing of EEG data, in which Z-score normalisation is applied to the scalp EEG data to eliminate outliers, values that deviate significantly from the mean. This step is crucial to avoid false detection of epilepsy events due to irregular data points. Different methods, such as Hamming, Hanning, Blackman, and Kaiser windows, can be used for real-time seizure detection. The main criteria are based on (i) side-lobe attenuation (artifact rejection), (ii) main-lobe width (frequency resolution vs. latency), (iii) effective window length (response time and memory), and (iv) computational load (multiplies per frame). In settings prioritising rapid alarm, Hamming and Hanning offer favourable latency with moderate attenuation, whereas Kaiser, with a β value, provides stronger suppression of out-of-band energy at the cost of slightly increased effective length.

A Kaiser window is used to manage signal leakage during filtering. Different values of β (beta) are considered (4, 6, and 8) to observe the effects on the frequency response. The sensitivity of the Kaiser window to the β parameter is important, as increasing β reduces spectral leakage and enhances side-lobe attenuation but simultaneously widens the main lobe and increases effective window length, raising group delay and computational burden. The specified β values balanced artifact suppression with response time to lower β reduced delay, but admitted more broadband noise. The value of a higher β improves the specificity at the expense of latency and multiplies per frame. The implementation on resource-limited hardware should consider β -dependent memory footprint and fixed-point scaling. To remove noise, a notch filter or band stop filter is applied, which is especially useful for eliminating power line interference. A high-order filter (order = 776) is used, indicating a very sharp and selective filtering process to isolate desired frequency bands. The Kaiser filter order ensures consistent and strong suppression of power-line noise across all EEG recordings, as the current evaluation emphasises frequency-domain parameters. The EEG data is sampled at 1000 Hz, meaning 1000 samples per second, which allows for capturing detailed temporal patterns in brain activity. Statistical techniques like mean, mode, median, standard deviation, and variance are used to summarize the EEG signal into measurable features for analysis. A machine learning neural network model is employed to classify whether the EEG signal indicates epileptic seizures or normal brain activity. The system produces the filtered EEG waveform, extracted features, and the classification result indicating seizure presence or absence. The performance metrics are evaluated, including relative leakage factor, Relative side attenuation, and main lobe width, with the classification metrics: accuracy, sensitivity, specificity, and precision.

EEG data collection: EEG data is collected online from physionet.org (<https://archive.physionet.org/cgi-bin/atm/ATM>). Raw EEG scalp data is downloaded, containing noise. Table 1 presents data categorized by age for both female and male patients, encompassing those who are

epileptic and non-epileptic [54].

Each case consists of 23 rows and 255 columns, where 255 represents the samples per row, and 23 corresponds to the channels of the acquired EEG signal. Thus, each case contains a total of $23 \times 255 = 5865$ samples. Since the total number of cases is six, the overall number of samples used in this study is $5865 \times 6 = 35,190$. The collected dataset includes one epileptic female, one non-epileptic female, and four male subjects, among whom two are epileptic, and two are non-epileptic.

3.1. Preprocessing of signals

The preprocessing of the raw EEG signal is conducted using the Z-score normalisation method in MATLAB, which eliminates outliers that deviate significantly from the mean. Eliminating these outliers prevents erroneous identification of epilepsy [55]. Z-score normalisation stabilises feature scales across segments and subjects, leading to more consistent decision thresholds and improved classifier robustness, particularly in cross-subject validation, and reduces threshold sensitivity, while filtering improves SNR and reduces false alarms from broadband artifacts. Ablation compares metrics with and without normalisation and filtering.

3.2. Kaiser windowing technique

The Kaiser window maximises the power ratio between the main and side lobes. The parameter β controls the sidelobe height. For a given β , the height of the sidelobe is fixed with the window length. As β grows, the height of the sidelobe drops as the breadth of the main lobe increases.

The Kaiser window function is expressed by

$$W_k(n) = \frac{I_0(\beta)}{\alpha} \quad \text{for } |n| \leq \frac{(M-1)}{2}$$

$$W_k(n) = 0 \quad \text{Otherwise}$$

Where β is the shape-controlling parameter (trade-off between side lobe attenuation and main lobe width). M is the window's length or the number of points. The symmetry range of the window is defined by α , where $(\alpha) = M/2$, and $I_0(\beta)$ is a Bessel function.

Notch Filter: The notch filter using the Kaiser windowing technique is used for filtering the noise signal of 50-60 Hz in the EEG signal [54, 56]. To eliminate unwanted frequency components, such as the typical power line noise in the 49–61 Hz range, a band-stop filter is applied using a windowed FIR (Kaiser) design. The frequencies are first normalized for the FIR filter design. The filter function is then used to apply the filter, which removes the undesirable frequencies through convolution. The resultant signal is a cleaner EEG signal with significantly reduced noise.

The frequency that a notch filter is designed to eliminate, such as 50 Hz or 60 Hz (common powerline noise), is called the center frequency (f_0). The bandwidth (BW) refers to the range of frequencies around (f_0) that are attenuated. A narrow bandwidth is preferred to minimize the distortion of nearby signals. The sharpness of the notch is determined by the quality factor (Q), defined as

$$Q = \frac{f_0}{BW}$$

A higher Q results in a sharper and narrower notch. Attenuation (Depth) refers to the level of suppression at the center frequency, typically measured in decibels (dB). A deeper notch is necessary when the interference is strong. To balance attenuation, transition width, and computational cost, a filter order of 776 is chosen. This value is suitable for the effective processing of EEG signals.

- Sampling frequency $F_s = 1000$ Hz
- Notch frequency $f_0 = 50$ Hz

Table 1

Patient details with age and gender wise.

Cases	Age(years)	Gender
Patient1	11	Female
Patient2	11	Female
Patient3	15	Male
Patient 4	15	Male
Patient 5	22	Male
Patient 6	22	Male

- Filter order $N = 776$

The relationship used to calculate the transition bandwidth (Δf) is

$$N = \frac{A - 8}{2.285 \times \Delta f}$$

$$\Delta f = \frac{A - 8}{2.285 \times N} = \frac{A - 8}{2.285 \times 776}$$

Assuming an attenuation $A = 60$ dB

$$\Delta f = \frac{52}{1773.96} = 0.0293$$

In Hz,

$$\Delta f.F_s = 0.0293 \times 1000 = 29.3 \text{ Hz}$$

This means the total transition bandwidth is approximately 29.3 Hz or ± 14.65 Hz around the notch center frequency of 50 Hz. Thus, the filter attenuates frequencies from approximately 35.35 Hz to 64.65 Hz.

Feature Extraction: The extracted features include minimum, maximum, mean, standard deviation, kurtosis, and skewness. These statistical measures were computed to capture the essential characteristics of the EEG signals. The following equations represent the mathematical expressions for mean, variance, skewness, and kurtosis.

Mean: Mean indicates the average signal amplitude of the EEG data and reflects its general baseline behaviour.

Mode: Mode refers to the amplitude value that appears most frequently within the EEG signal.

Standard Deviation (SD): Standard deviation expresses the degree of variation in EEG signal amplitudes relative to the mean value.

Variance: Variance quantifies the average squared deviation of EEG signal amplitudes from their mean, reflecting the overall signal power and variability.

Skewness: Skewness measures the degree of asymmetry present in the amplitude distribution of the EEG signal.

Kurtosis: Kurtosis characterises the sharpness or flatness of the EEG signal's amplitude distribution.

$$\text{Mean } (\mu) = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

$$\text{Variance } (\sigma^2) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

$$\text{Skewness} = \frac{1}{N} \sum_{i=1}^N \frac{(x_i - \mu)^3}{\sigma^3}$$

$$\text{Kurtosis} = \frac{1}{N} \sum_{i=1}^N \frac{(x_i - \mu)^4}{\sigma^4}$$

Here,

- x_i : i th sample of the signal,
- N : Total number of samples,
- μ : Mean of the signal,
- σ : Standard deviation of the signal,
- σ^2 : Variance of the signal,
- i : Sample index, where $i = 1, 2, \dots, N$.

3.3. Classification using a neural network machine learning algorithm

Mathematical models that imitate the structure and function of

neural networks in the human brain are known as neural networks in the computer world, sometimes called artificial neural networks. We use an ANN classification machine learning [57,58] to determine if an EEG signal is epileptic or not based on its filtered, feature-extracted output. Two hidden layers and a single output characterize this artificial neural network [59,60]. An epileptic patient is indicated by an output of '1', while a non-epileptic individual is indicated by an output of '0'. An ANN is a supervised machine-learning model inspired by the structure and functioning of the human brain. Its strong pattern recognition capabilities make it well-suited for applications such as disease detection, image recognition, and EEG signal classification. In supervised learning, the model is trained using labeled data, meaning both the input and the corresponding output are known during training. For EEG analysis, filtered signals are obtained from electrodes placed at positions such as FP1, FP7, P3-O1, and others. From these signals, relevant features are extracted, and an ANN is then used to classify these features, for example, distinguishing between epileptic and non-epileptic states. An ANN is composed of multiple layers:

- **Input Layer:** Receives the extracted features from the EEG signals.
- **Hidden Layer:** Performs computations. In each hidden layer, neurons compute a weighted sum of the inputs, followed by the application of an activation function. Consider the following inputs: $x_1, x_2, x_3, \dots, x_n$. These inputs are processed using the associated weights $w_1, w_2, w_3, \dots, w_n$, and the bias input 'b' [61]. The equation is used to express neuron processing.

$$x = \sum_{i=1}^n x_i w_i + \text{Bias } (b)$$

- **Output Layer:** Produces the final classification result, for example, class 0 for a normal EEG signal and class 1 for an epileptic signal.

4. Results and discussions

The raw EEG data is first loaded from [Physionet.org](https://physionet.org). Physionet.org is a reliable, open-source platform that provides high-quality physiological datasets for research, education, and development. EEG data is frequently sourced from this site. Specifically, Physionet offers:

- Real EEG recordings were collected from medical research institutions, hospitals, and clinical studies.
- Open-access data makes it ideal for researchers, developers, and students.
- A wide range of EEG datasets related to neurodegenerative diseases, sleep studies, epilepsy, and brain-computer interface research.

The EEG signals are acquired from various electrode positions of the brain. The noise from the EEG scalp data is removed with the help of a Kaiser window FIR filter notch filter. The software used for this is MATLAB2023a and the Filter Design Analysis (FDA) tool. The EEG signal shown in Fig. 2 presents the brain's electrical activity over 10 s. Time (seconds) is represented on the x-axis, while amplitude (μV), ranging from $-400 \mu V$ to $+300 \mu V$, is displayed on the y-axis.

- **0–2 s:** The signal oscillates moderately around the baseline ($0 \mu V$). The amplitude typically stays within $\pm 100 \mu V$, characteristic of alpha or beta EEG rhythms. A few spikes are visible, likely caused by minor muscle artifacts or normal brain activity.
- **2–4 s:** Around 3–3.5 s, the signal becomes denser, with more frequent and sharper peaks. This may be due to increased cognitive activity, low-level noise, or muscle movement. The amplitude in this region reaches around $\pm 200 \mu V$.
- **4–6 s:** The activity grows more irregular and intense, with larger amplitude swings, occasionally exceeding $-300 \mu V$. Around the 5-s

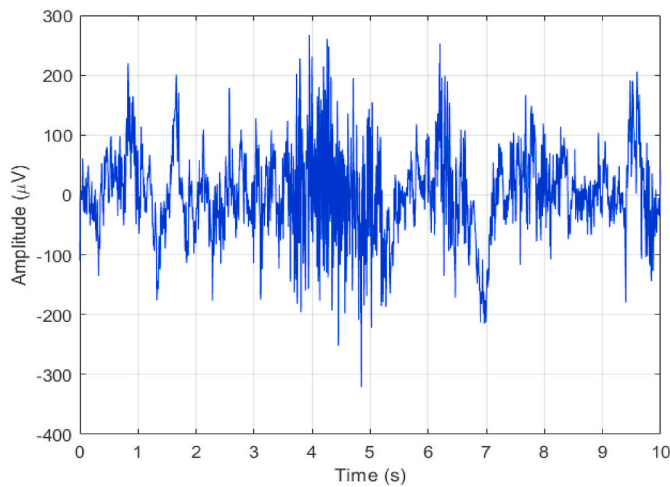


Fig. 2. EEG signal waveform.

mark, a noticeable peak suggests a potential artifact or a sudden surge in brain activity. Some high-frequency components are present, possibly indicating gamma band activity or muscular interference.

- **6–8 s:** Although the amplitude slightly decreases, the signal remains highly active, possibly indicating sustained cognitive engagement or atypical EEG patterns. A dip nearing $-250 \mu\text{V}$ around 7 s may represent a transient EEG burst.
- **8–10 s:** A few stronger bursts are observed near 9.5 s. The signal continues to fluctuate with moderate intensity, and dense, high-frequency components persist until the end. The amplitude remains within the $\pm 200 \mu\text{V}$ range.

Fig. 3(a)–(w) shows the EEG signal taken from the scalp for different channels and from different electrode positions. F denotes the frontal lobe (anterior section of the brain), whereas P signifies the parietal lobe (region superior to the neck). T denotes the temporal region (located just posterior to the ear), O signifies the occipital area (the posterior aspect of the skull), and C represents the central region of the brain. Z represents the midline center or zero offset; odd numbers are processed in the left hemisphere of the brain, while even numbers are processed in the right hemisphere. FP₁ indicates an EEG electrode positioned on the left frontal forehead. FP₇ EEG electrode positioned on the left temporal forehead. FP₁–FP₇: This channel measures the electrical activity between the brain's lateral-left (FP₇) and frontal-left (FP₁) regions. It depicts the fluctuations in neuronal activity within that region, especially in the left frontal and temporal lobes. This channel records activity from the left lateral frontal to mid-temporal regions of the brain. When analyzing electrical activity that localizes or propagates inside the left temporal lobe, is highly beneficial. T₇–P₇: This bipolar channel indicates electrical activity along the left lateral temporal to parietal pathway. P₇–O₁ records cerebral activity from the left parietal to the left occipital area. The FP₁–F₃ EEG channel indicates the electrical activity between the left prefrontal cortex (FP₁) and the left frontal cortex (F₃). The bipolar channel records electrical activity transitioning from the left frontal cortex to the left central (motor) cortex. P₃–O₁ highlights electrical activity between the left parietal and left occipital areas of the brain. FP₂–F₄: This bipolar channel records the electrical activity between the right prefrontal cortex and the right frontal cortex. F₄–C₄: This bipolar channel captures electrical activity from the right frontal lobe, associated with planning, decision-making, and attention, to the right central region, which governs motor function, especially on the left side of the body. C₄–P₄: It indicates neuronal activity between the right central and right parietal regions of the brain.

The processing of the EEG signal is done in MATLAB, and 50 to 60 Hz

noise from the raw EEG scalp data is removed with the help of the FIR Kaiser window notch filter the order of the filter is 776, and the sampling frequency used is 1000 Hz. The waveform for Raw EEG scalp data and filtered EEG scalp data is shown in Fig. 4. The Magnitude response in (dB) and round-off Noise spectrum of the Notch FIR filter with the Kaiser window technique are shown in Fig. 5. The step response of the band-reject FIR filter with the Kaiser window technique is shown in Fig. 6. The impulse response of the notch FIR filter with the Kaiser window technique is shown in Fig. 7. The Magnitude response of the notch FIR filter with the Kaiser window technique is shown in Fig. 8. Table 2 shows the parameter analysis for the Kaiser window with ($\beta = 4$, $\beta = 6$, and $\beta = 8$). In the context of filters, the phrase ‘leakage factor’ usually refers to the amount of signal or spectral leakage that happens outside of the filter's main passband. Leakage can cause undesired effects such as increased noise or interference from adjacent frequency components. Ideally, the leakage factor should be as small as possible. This means that the filter successfully restricts the signal to its targeted passband while minimizing signal leakage into adjacent frequency ranges.

The fundamental goal of a notch filter is to achieve significant attenuation within the stopband, which essentially eliminates or greatly reduces sounds within the defined frequency range. As a result, while constructing a band-reject filter, we normally aim for higher relative side attenuation inside the stopband (the notch frequency range) to effectively reject undesired frequencies while minimizing attenuation within the passbands. This guarantees that the filter selectively removes frequencies within the given range while having no meaningful effect on signals outside of that range. The main lobe width should be very small or near zero so that the filter is highly selective. Parameter analysis for the FIR Kaiser window with a notch filter. The Kaiser ($\beta = 12$) window has a 0% relative leakage factor, relative side lobe attenuation is -89.9 dB and, which is higher side lobe attenuation in comparison to other Kaiser ($\beta = 4$) windows, Kaiser ($\beta = 6$) window, and Kaiser ($\beta = 8$) window. For highly selective filters main lobe width should be nearer to 0. FIR Kaiser window ($\beta = 12$) with notch filter main lobe width value is 0.002686, which is nearer to zero. So, the Parameter analysis for the FIR Kaiser window with notch filter shows that the Kaiser ($\beta = 12$) window is a good window for filtering Raw EEG scalp data.

The frequency at which the filter significantly attenuates the signal is indicated by a dramatic dip in the magnitude response at the notch frequency. The effectiveness of the filter in eliminating that frequency component is indicated by the depth of this notch. There is a dramatic dip in the magnitude response in the frequency range from 50 to 60 Hz. This shows that at this frequency, the filter significantly attenuates the signals.

The filter's response to an abrupt shift in the input signal is demonstrated by the step response. A good filter should ideally not cause significant overshoot or ringing and should soon settle into a stable state. The step response of the band-reject FIR filter with the Kaiser window technique shows that it does not have significant overshoot, and it also soon settles to a stable state.

By windowing the ideal impulse response using a Kaiser window, the band-reject FIR filter's impulse response using a Kaiser window technique can be expressed as a finite sequence. Through the Kaiser window's form, side-lobe levels are controlled while the filter is guaranteed to satisfy the required frequency characteristics. The impulse response of a band reject filter will have a characteristic shape, with near-zero values elsewhere (which represent the stopband) and significant values around the center (which represent the passbands). A band-reject FIR filter created with the Kaiser window technique's magnitude response shows how different frequency components of an input signal are passed through or attenuated. It displays the filter's gain (or attenuation) at different frequencies.

Table 3 shows a comparison of different windowing methods commonly used in EEG signal processing, focusing on their performance in terms of relative leakage factor, side lobe attenuation, and main lobe width (in mHz).

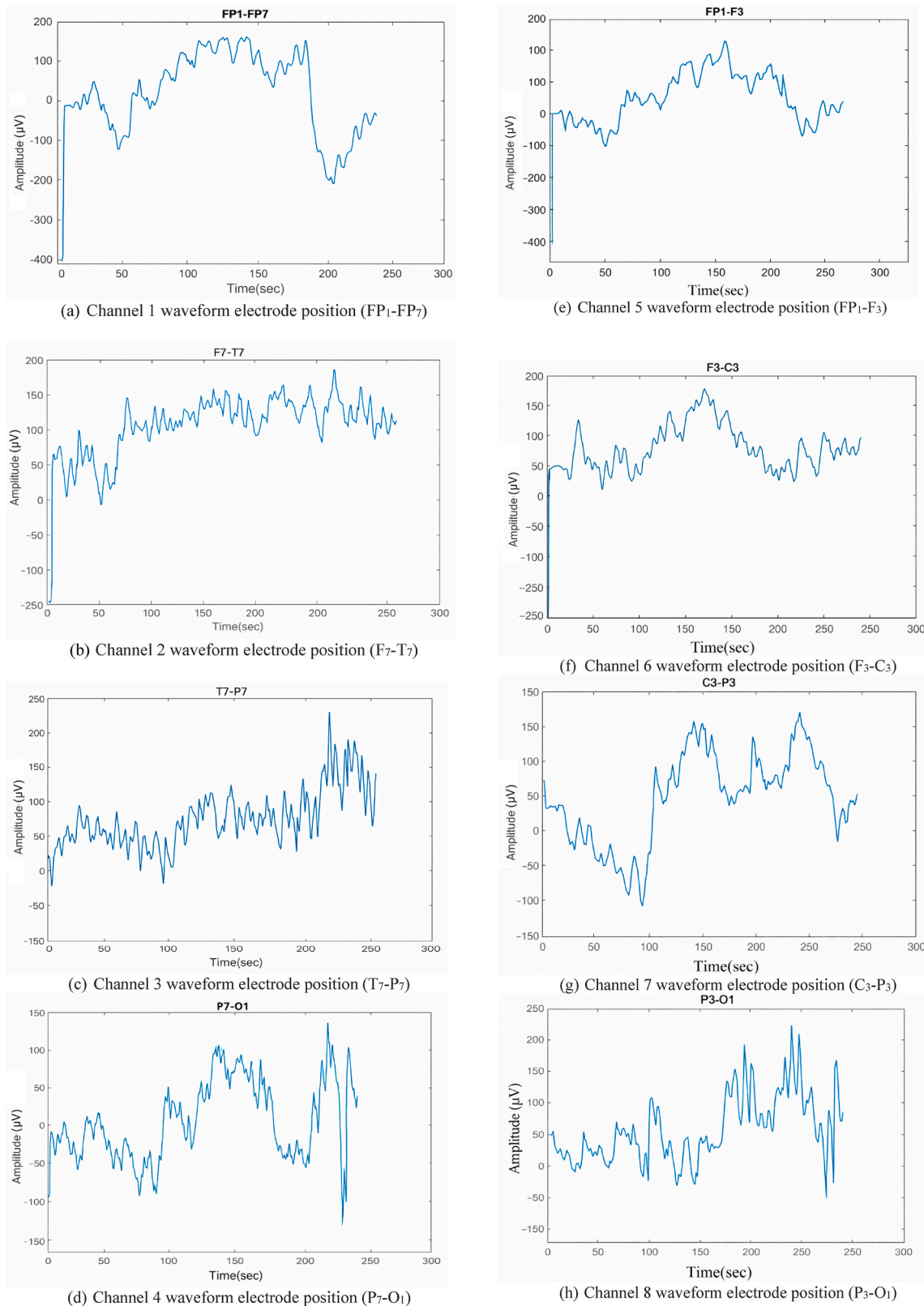


Fig. 3. All channels and electrodes signal waveforms.

- The Rectangular window has the highest leakage (9.151%) and lowest side lobe attenuation (13.3 dB), though it provides a narrow main lobe width of 0.054688 mHz, which can lead to poor frequency isolation.
- The Hamming window offers very low leakage (0.04%) and good side lobe suppression (41.7 dB), with a slightly wider main lobe of 0.085938 mHz, making it suitable for cleaner frequency representation.

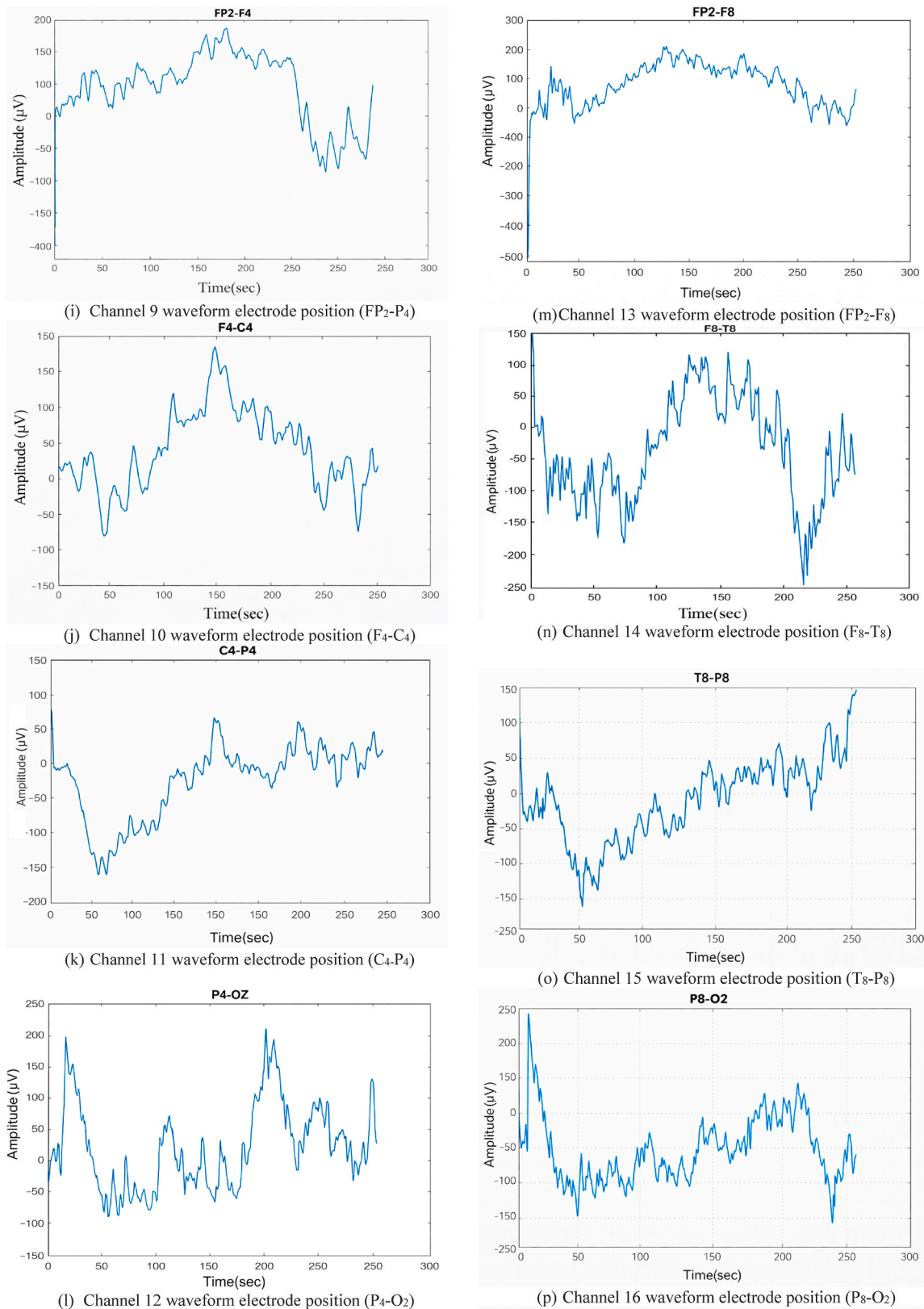


Fig. 3. (continued).

- The Bartlett window has moderate values across all parameters, with 0.28% leakage, 26.5 dB attenuation, and the same main lobe width as the Hamming window.
- The Hanning window performs similarly, with 0.05% leakage, 31.5 dB side lobe attenuation, and a slightly wider main lobe of 0.093750 mHz, offering a balanced option.
- The Kaiser window stands out with zero leakage, exceptional side lobe attenuation (−89.9 dB), and the narrowest main lobe width

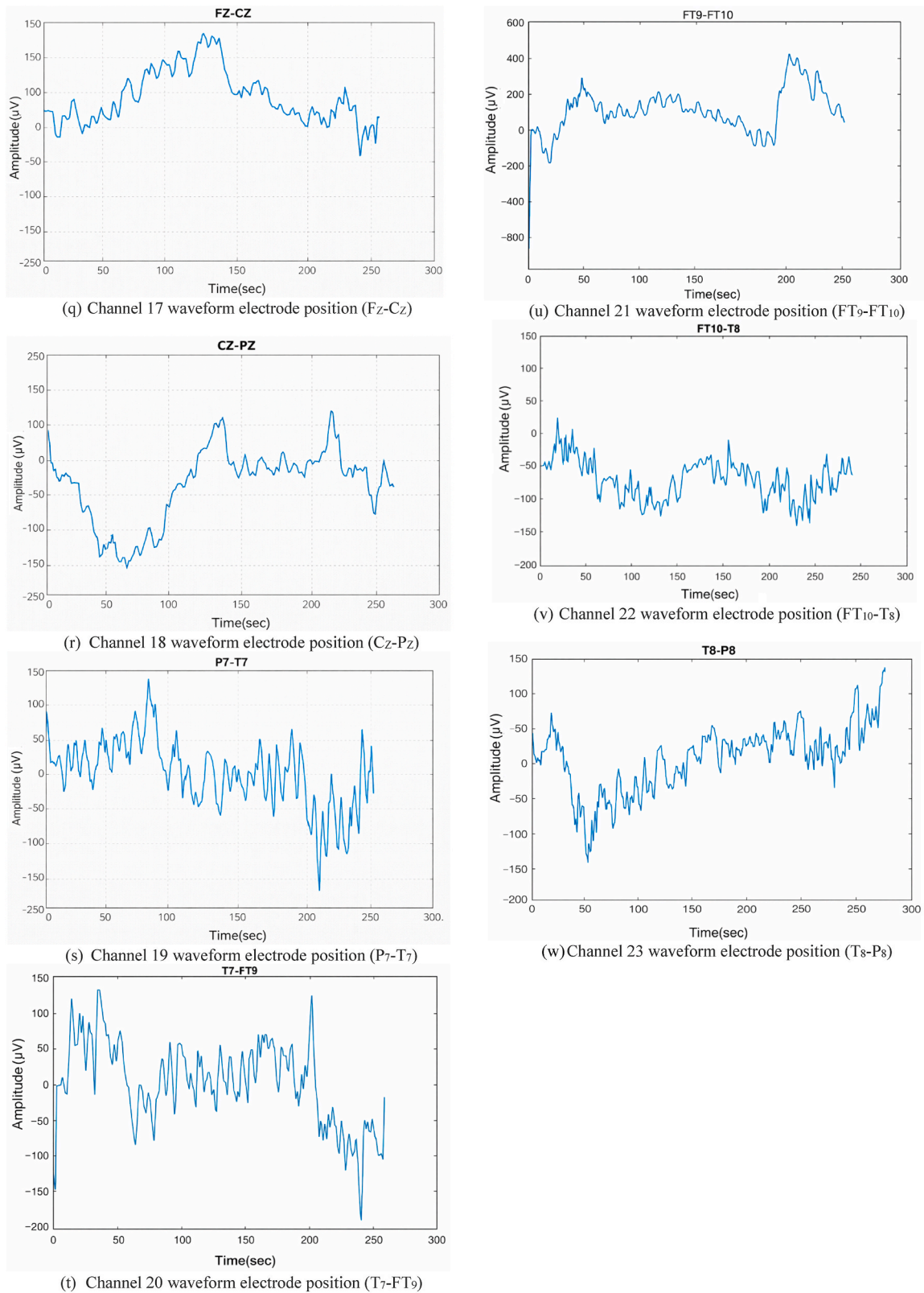


Fig. 3. (continued).

(0.002686 mHz), making it the most effective for minimizing noise and improving frequency resolution in EEG analysis.

The filtered EEG signal, processed using the Kaiser window technique, yields six statistical features: mean, mode, median, variance,

kurtosis, and skewness, which are extracted using MATLAB 2023a. These extracted features are subsequently supplied to a machine learning classifier, specifically an ANN, for automated epileptic seizure detection. The distributions of the retrieved features, including variance, standard deviation, mean, mode, and median, are illustrated in Fig. 9

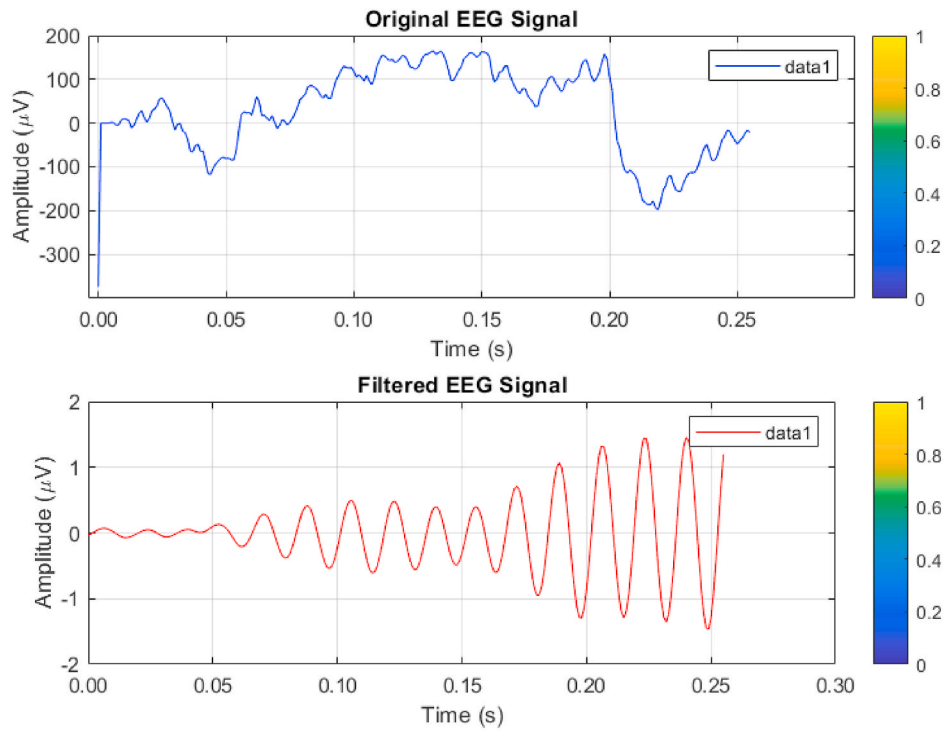


Fig. 4. Original EEG signal and filtered EEG signal of band reject FIR filter with the Kaiser window technique.

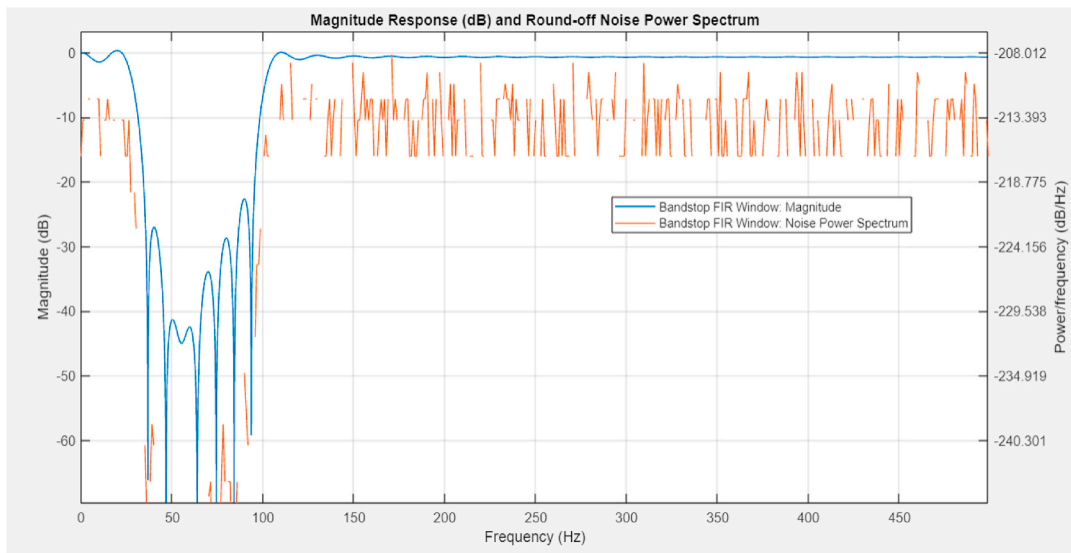


Fig. 5. Magnitude response (dB) and round-off noise spectrum of band-reject FIR filter with Kaiser window technique.

(a)–(g), respectively.

4.1. The histogram of mode values

- The highest frequency is near zero, with around 90 values clustered in the bin closest to 0.
- 30 values fall within the bin range $[-0.03, -0.02]$.
- A smaller count of 7 values appears in the bin range $[-0.07, 0.06]$.

This suggests that most mode values are concentrated near zero, with gradually decreasing frequencies toward more negative values.

4.2. The histogram of mean values

- A dominant peak shows 120 values in the bin range $[-0.05, 0] \times 10^{-3}$
- A single outlier value falls in the bin range $[-1.25, -1.2] \times 10^{-3}$

This indicates that most data points are clustered very close to zero, with one significantly lower value as an anomaly.

4.3. The histogram of median values

- Displays a symmetric distribution centred around zero.
- The highest frequency (75 values) lies in the bin range $[-0.0005, 0]$.
- A secondary peak (24 values) is in the $[-0.001, -0.0005]$ bin

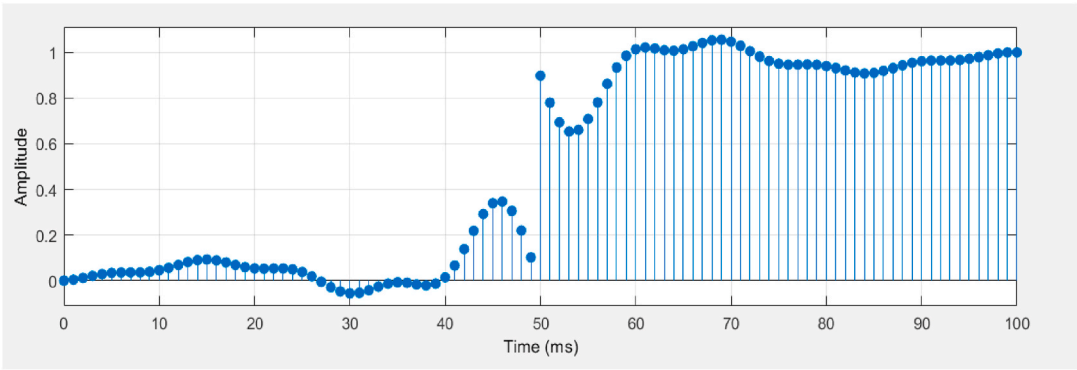


Fig. 6. Step response of band-reject FIR filter with Kaiser window technique.

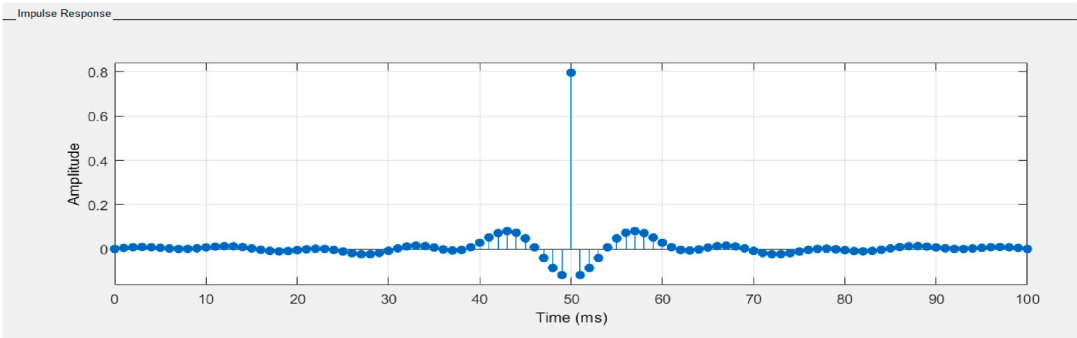


Fig. 7. Impulse response of band-reject FIR filter with Kaiser window technique.

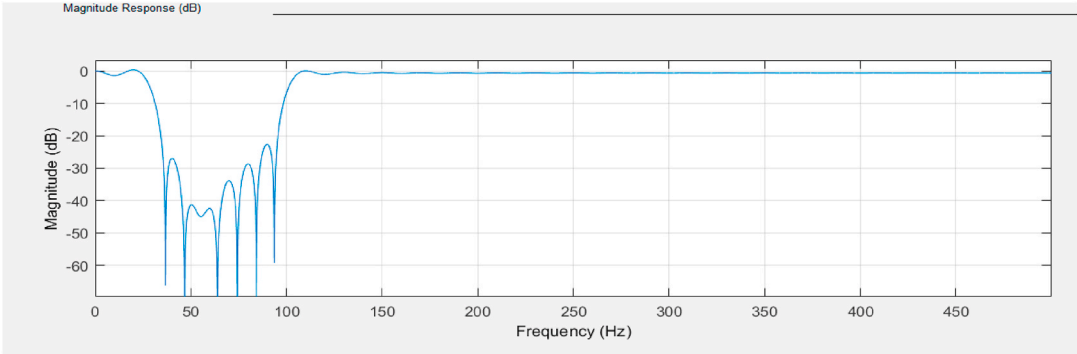


Fig. 8. Magnitude response of band-reject FIR filter with Kaiser window technique.

Table 2
Parameter analysis for the FIR Kaiser window with a notch filter.

Window used	Filter	Relative leakage factor (%)	Relative side lobe attenuation (dB)	Mainlobe width (mHz)
Kaiser ($\beta = 4$)	Notch filter	0.1	-30.0	0 0.001465
Kaiser ($\beta = 6$)	Notch filter	0.0	-43.0	0.001709
Kaiser ($\beta = 8$)	Notch filter	0.0	-58.7	0.001953
Kaiser ($\beta = 12$)	Notch filter	0.0	-89.9	0.002686

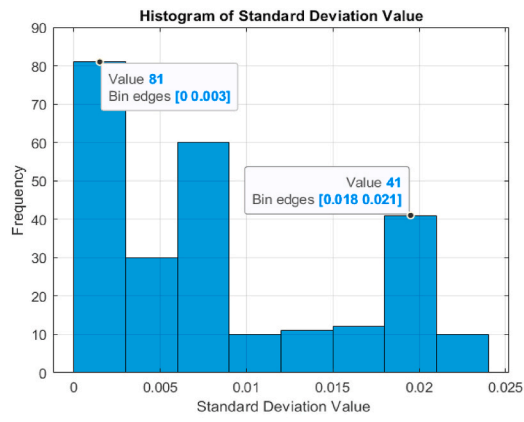
The values are tightly clustered near zero, indicating a central tendency with very low skewness and minimal median variation across the dataset.

Table 3
Comparison with other windows for EEG.

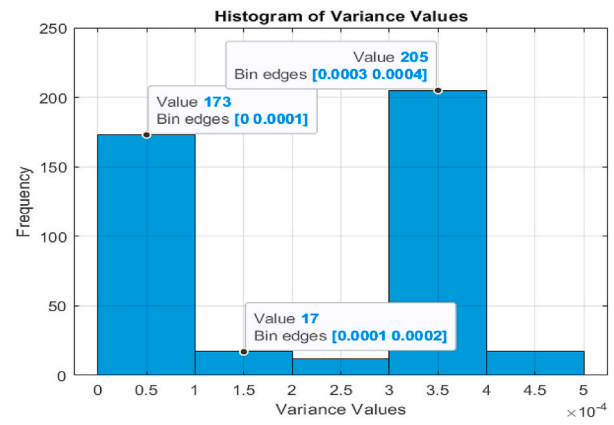
Window method	Relative leakage factor %	Relative Side lobe attenuation (dB)	Main lobe width (mHz)
Rectangular	9.151	13.3	0.054688
Hamming	0.04	41.7	0.085938
Barlett	0.28	26.5	0.085938
Hanning	0.05	31.5	0.093750
Kaiser	0.0	-89.9	0.002686

4.4. The histogram of variance values

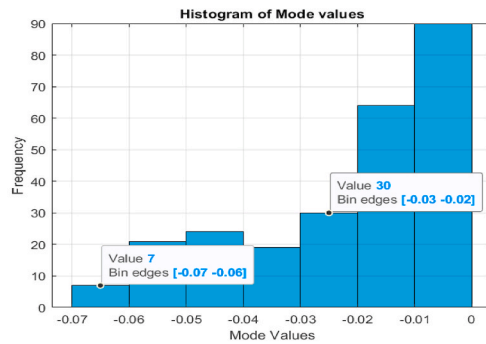
- It shows three prominent groups. 205 values fall in the bin [0.0003, 0.0004], indicating high clustering.
- 173 values are in the [0, 0.0001] range, also significantly represented.



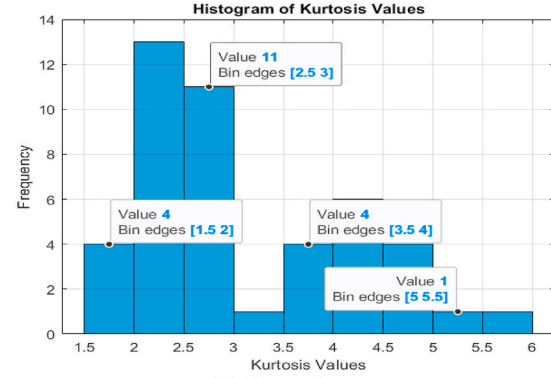
(a) Standard deviation



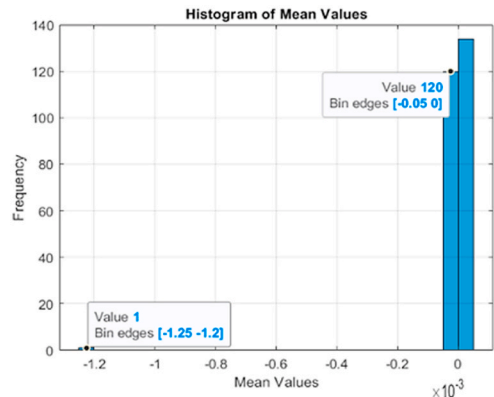
(e) Variance



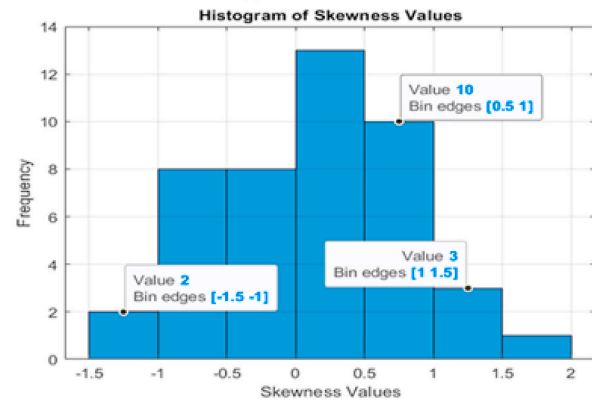
(b) Mode



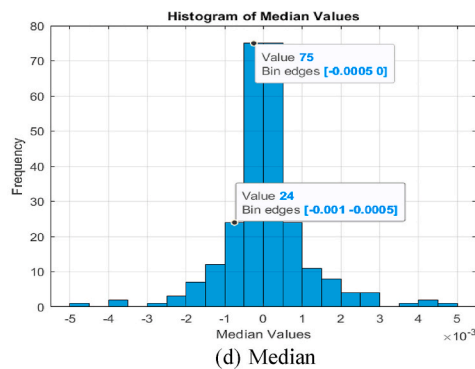
(f) Kurtosis



(c) Mean



(g) Skewness



(d) Median

Fig. 9. Retrieved features as histogram values.

- A smaller peak with 17 values appears in [0.0001, 0.0002].

Overall, the variance values are mostly concentrated at the lower and higher ends, suggesting a bimodal distribution in data spread.

4.5. The histogram of kurtosis

- The highest frequency (11 values) lies in the [2.5, 3] bin range.
- Moderate counts include 4 values each in [1.5, 2] and [3.5, 4] bins.
- A rare extreme (1 value) appears in [5, 5.5].

This suggests that most data distributions are mesokurtic (around kurtosis = 3), with some skew toward leptokurtic (peaked) and platykurtic (flat) tails.

4.6. The histogram of skewness

- The most frequent bin is [0, 0.5], containing 13 values, indicating slight right skewness in most data.
- 10 values lie in the [0.5, 1] range, suggesting moderate right skew.
- Few values fall in negative skewness bins like [-1.5, -1] (2 values), showing that left-skewed data is rare.

High variance denotes EEG signal segments with stronger power or amplitude fluctuations. High-variance features could be artifacts like eye blinks or muscle movements, or they could be brain states linked to intense neuronal activity. A low variance type of signal indicates greater consistency or stability and may be indicative of more stable brain states, such as resting or relaxed states. The extent to which the variation of the EEG signal characteristics diverges from the mean is evident when we calculate the standard deviation from this histogram. The mean of the extracted features in an EEG feature-extracted histogram indicates the average variance or central tendency of the feature values throughout the dataset. This helps to clarify the general degree of variability in the EEG signals following feature extraction. The stratified subject-exclusive K-fold cross-validation is performed with $K = 5$. This setting ensures at least 6 subjects per class in each test fold, given our cohort size, improving the reliability of fold-level metrics. Inner 3-fold CV on the training split tunes hyperparameters. We report mean $\pm$ SD across folds and 95% CIs. If an external dataset is available, we report results once on that set, maintaining strict subject exclusivity. The extent to which the variation of the EEG signal characteristics diverges from the mean is evident when we calculate the standard deviation from this histogram. The predominant value of the extracted features (variance or other statistical attributes) in a feature-extracted histogram of EEG data is referred to as the mode. The median, defined as the value that divides the dataset into two halves, is the central value of the feature distribution in a feature-extracted histogram of EEG data. The median is a valuable measure of central tendency, particularly in the presence of skewed or noisy data, as it is less influenced by extreme values (outliers) compared to the mean. Following feature extraction, the data is provided to the ANN for the classification of EEG signals as either epileptic or non-epileptic. 80% of the data is allocated for training purposes, whereas 20% is designated for testing purposes. The classification accuracy achieved with the neural network is 99.6%. The confusion matrix is illustrated in Fig. 10.

Fig. 11 shows that the TPR detection is 99.3 %, and the false negative rate is 0.7%. Based on the confusion matrix, the sensitivity, specificity, and precision can be calculated as follows: Table 4 lists the performance measurement parameters of classification. From the confusion matrix,

True Positives (TP): 138,
False Positives (FP): 0,
True Negatives (TN): 116,
False Negatives (FN): 1.

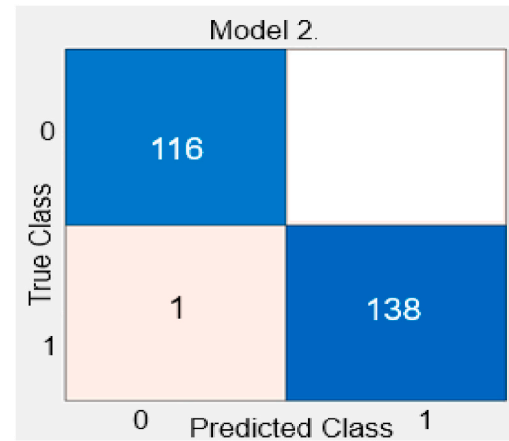


Fig. 10. Confusion matrix.

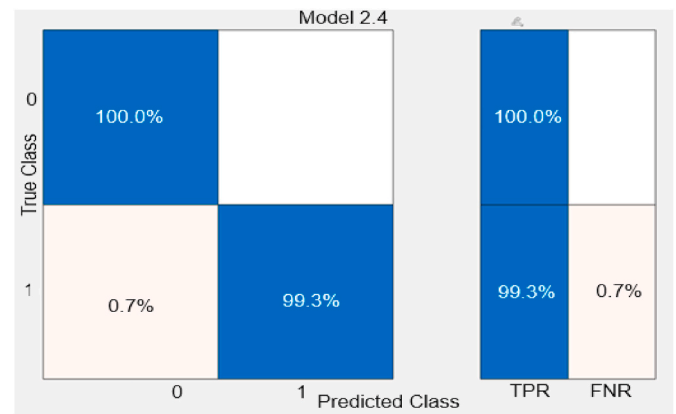


Fig. 11. True positive rate (TPR) and False negative rate (FNR) Matrix.

The correlation matrix presented in Fig. 12 illustrates the model's efficacy in distinguishing between "highly epileptic" and "non-epileptic" conditions, as seen by the cells where the actual label (y-axis) aligns with the predicted label (x-axis). Darker diagonal cells indicate higher accuracy for the classification of non-epileptic versus epileptic conditions. A dark diagonal cell in class 1 (epileptic) signifies that the model has great accuracy in identifying epileptic patients. A dark diagonal would signify high precision in identifying non-epileptic patients, or class 0 (non-epileptic). The correlation coefficients between the columns (i.e., the 20 EEG channels) were calculated to form a 20×20 correlation matrix. Each cell at position (i,j) in the matrix represents the correlation coefficient between channels i and j. This analysis helps in identifying the relationships between different EEG channels. The diagonal of the matrix, which represents the correlation of each channel with itself, is highlighted in dark blue and has a value of 1, indicating perfect correlation.

The False Positive Rate (X-axis) indicates the proportion of negative samples misclassified as positive. The True Positive Rate (Y-axis) indicates the proportion of positive samples accurately identified as positive, commonly referred to as sensitivity or recall. The Area Under the Curve (AUC) for both classes (0 and 1) is 0.9852, indicating excellent performance as depicted in Fig. 13. The AUC varies from 1, indicating excellent classification, to 0.5, which signifies no discrimination. An elevated AUC signifies the model's strong capability in distinguishing between classes, such as epileptic and non-epileptic.

Despite strong performance, the model may overfit subject-specific patterns, be sensitive to class balance and montage differences, and degrade under distribution shift (device, protocol, clinical artifacts). To mitigate these risks, we employ subject-wise CV, regularisation,

Table 4
Performance measurement parameters of classification.

Sensitivity = $TP/(TP + TN) = 138/(138 + 1) = 138/139 = 0.993$ or 99.3%
Specificity = $TN/(TN + FP) = 116/(116 + 0) = 116/116 = 100.0\%$
Precision = $TP/(FP + TP) = 138/(0 + 138) = 138/138 = 100.0\%$
Accuracy = $(TP + TN)/(TP + TN + FP + FN) = (138 + 116)/(138 + 116 + 0 + 1) = 99.6\%$

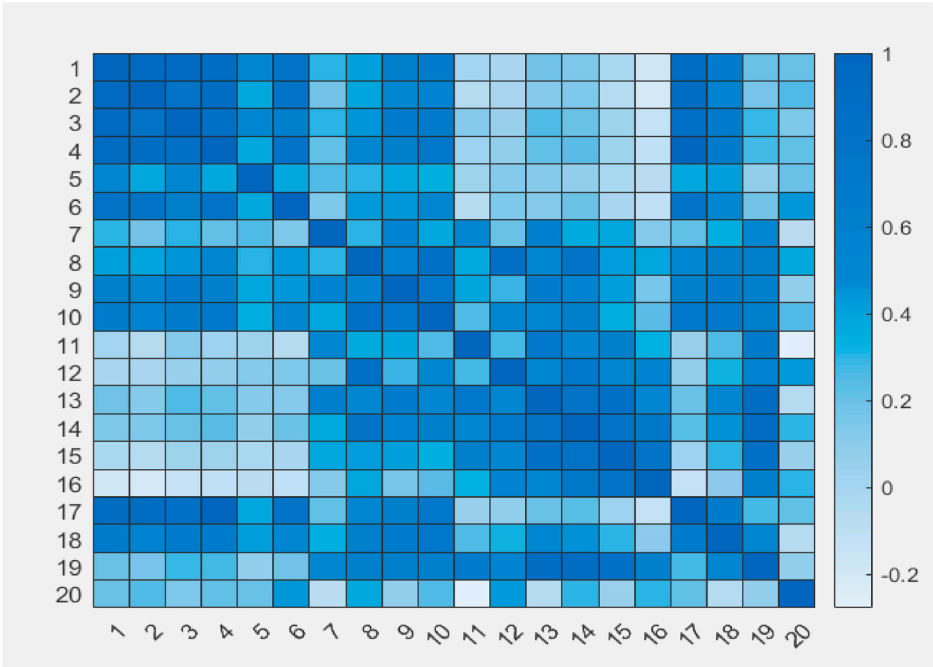


Fig. 12. Correlation matrix.

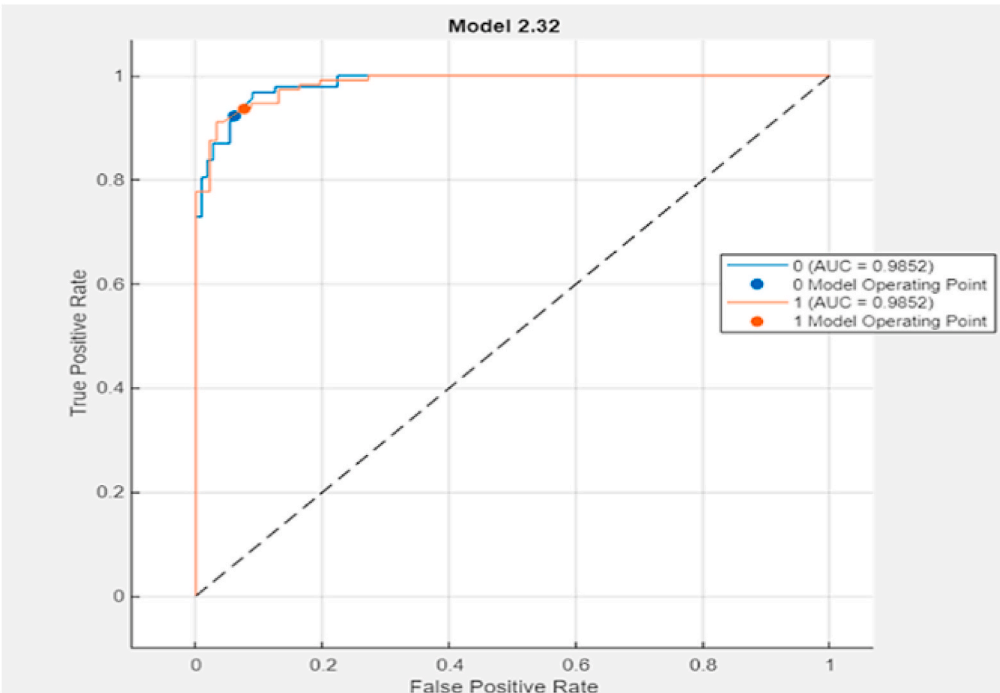


Fig. 13. AUC curve.

augmentation with artifact simulations, and calibration/thresholding on validation data. Clinical deployment can warrant prospective validation and periodic recalibration.

5. Conclusion

The EEG signal is obtained by applying a notch filter using the Kaiser windowing technique response of the band-reject FIR filter using the Kaiser window technique is analyzed in the MATLAB 2023a software, with the FDA tool. The filter has a length of 775 and an order of 776. The frequency at which samples are taken is 1000Hz. Performing parameter analysis for a notch filter implemented using the FIR Kaiser window. The comparative study reveals that the Kaiser ($\beta = 12$) window has a Relative Leakage Factor of 0% and a higher Relative Side Lobe Attenuation of -89.9 dB compared to the Kaiser ($\beta = 4$), Kaiser ($\beta = 6$), and Kaiser ($\beta = 8$) windows. To have a high selectivity filter, the main lobe width should be closer to 0. The FIR Kaiser window with a shape parameter (β) of 12 is used in conjunction with a notch filter. The main lobe width value of this window is 0.002686, which is very close to zero. The parameter study of the FIR Kaiser window with a notch filter demonstrates that the Kaiser window with a β value of 12 is highly effective for filtering raw EEG scalp data. The filtered EEG data is broken down into five statistical features: mean, mode, median, variance, and standard deviation. These are subsequently processed by the machine learning algorithm for categorization. 80% of the data is allocated for training, while 20% is reserved for testing. This study introduces a robust FIR-based EEG signal processing system using Kaiser window techniques, demonstrating the accuracy of notch filtering using a $\beta = 12$ window. The neural network classification and statistical feature extraction system detects epileptic signals 99.6% accurately. The results show that high-selectivity filters improve EEG categorization. Future research will focus on FPGA real-time implementation, deep learning-based temporal models, and multi-class and cross-dataset validation to prepare for clinically deployable, intelligent EEG diagnostic tools. The transition from offline signal analysis to real-time embedded deployment may introduce slight performance variation due to latency limits, coefficient quantisation, and memory or processing constraints. Considering these factors provides a more practical view of the reported filtering efficiency and underscores the need for on-device validation to confirm stability under real-time operating conditions.

The study acknowledges certain limitations that define the scope of the current work. Although the filtering method effectively maintains phase integrity through a linear-phase FIR Kaiser-based notch filter, the research does not include clinical-level validation of phase distortion effects. The discussion on phase preservation has been added, yet its impact on real clinical assessments remains unexplored. Furthermore, the work focuses solely on filtering, feature extraction, and classification for seizure detection, without extending to broader clinical trials or multi-centre evaluations. These aspects will be considered in future investigations to strengthen the practical applicability of the proposed approach.

Credit authors statement

Aruna Pant: Conceptualization, Methodology, Validation, Software, Writing – original draft.

Adesh Kumar: Methodology, Supervision, Resources, Project Administration, Writing – review & editing.

Funding

This research received no specific grant from any funding agency in the public and private sectors.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: We declare that no known competing financial interests or personal relationships that could have appeared to influence the work reported in

this paper.

Abbreviations

AUC	Area Under Curve
BCI	Brain Computer Interface
BW	Bandwidth
C	Central (EEG electrode region)
CNN	Convolutional Neural Network
CSD	Current Source Density
CV	Cross-Validation
ECG	Electrocardiogram
EEG	Electroencephalography
FDA	Filter Design and Analysis
FFT	Fast Fourier Transform
FIR	Finite Impulse Response
FN	False Negative
FNR	False Negative Rate
FP	False Positive
FP1/FP7/FP2/FP4	Fronto-polar EEG electrode positions
Fs	Sampling Frequency
FZ/CZ/PZ	Midline EEG electrode positions
HCI	Human–Computer Interaction
Hz	Hertz
IIR	Infinite Impulse Response
KNN	K-Nearest Neighbour
LDA	Linear Discriminant Analysis
LSTM	Long Short-Term Memory
M –	Window Length (number of FIR coefficients)
MATLAB	Matrix Laboratory software
MSE –	Mean Squared Error
μ V –	Microvolt
O –	Occipital (EEG region)
P –	Parietal (EEG region)
PSD –	Power Spectral Density
Q	Quality Factor (Notch filter sharpness)
RNN –	Recurrent Neural Network
SD –	Standard Deviation
SNR –	Signal-to-Noise Ratio
SMR	Sensorimotor Rhythm
SVM –	Support Vector Machine
T	Temporal (EEG region)
TN	True Negative
TP	True Positive
TPR	True Positive Rate
μ V	Microvolt
Z-score	Standard score normalisation method

Data availability

All data generated or analyzed during this study are included in this published article.

References

- [1] R. Sharma, H.K. Meena, Emerging trends in EEG signal processing: a systematic review, *SN Comput. Sci.* 5 (4) (2024) 1–14.
- [2] H. Aurlén, I.O. Gjerde, J.H. Aarseth, G. Eldøen, B. Karlsen, et al., EEG background activity described by a large, computerized database, *Clin. Neurophysiol.* 115 (2004) 665–673.
- [3] M.S. Maan, EEG waves classifier using wavelet transform and fourier transform, *Int. J. Biomed. Sci.* 1 (2007) 85–90.
- [4] A.S. Al-Fahoum, A.A. Al-Fraihat, Methods of EEG signal features extraction using linear analysis in frequency and time-frequency domains, *ISRN Neurosci.* 2014 (2014) 730218.
- [5] V. Janiukstyte, T.W. Owen, U.J. Chaudhary, B. Diehl, L. Lemieux, J.S. Duncan, P. N. Taylor, Normative brain mapping using scalp EEG and potential clinical application, *Sci. Rep.* 13 (1) (2023) 13442.

- [6] M. Lai, M. Demuru, A. Hillebrand, M. Fraschini, A comparison between the scalp and source-reconstructed EEG networks, *Sci. Rep.* 8 (1) (2018) 12269.
- [7] L. Dong, F. Li, Q. Liu, X. Wen, Y. Lai, P. Xu, D. Yao, MATLAB toolboxes for reference electrode standardization technique (REST) of scalp EEG, *Front. Neurosci.* 11 (2017) 601.
- [8] S. Wong, A. Simmons, J. Rivera-Villicana, S. Barnett, S. Sivathamboo, P. Perucca, T.J. O'Brien, EEG datasets for seizure detection and prediction—A review, *Epilepsia Open* 8 (2) (2023) 252–267.
- [9] J.W. John, *Medical Instrumentation Application and Design*, fourth ed., Wiley, 2010.
- [10] A. Kilcarslan, R.G. Grossman, J.L. Contreras-Vidal, A robust adaptive denoising framework for real-time artifact removal in scalp EEG measurements, *J. Neural. Eng.* 13 (2) (2016) 26013.
- [11] E. Niedermeyer, Ernst Niedermeyer, F.L. Silva, *Electroencephalography: Basic Principles, Clinical Applications and Related Fields*, Lippincott Williams & Wilkins, 2004.
- [12] N. Karaboga, B. Cetinkaya, Design of digital FIR filters using differential evolution algorithm, *Circ. Syst. Signal Process.* 25 (5) (2006) 649–660.
- [13] A. Miltiadous, K.D. Tzimourta, T. Afrantou, P. Ioannidis, N. Grigoriadis, D. G. Tsilikakis, A.T. Tzallas, A dataset of scalp EEG recordings of Alzheimer's disease, frontotemporal dementia and healthy subjects from routine EEG, *Data* 8 (6) (2023) 95.
- [14] K.D. Chinchkhede, G.S. Yadav, S.R. Hirekhan, D.R. Solanke, On the implementation of FIR Filter with various windows for enhancement of ECG signal, *Int. J. Eng. Sci. Technol.* 3 (3) (2011) 2031–2040.
- [15] D. Nhu, M. Janmohamed, A. Antonic-Baker, P. Perucca, T.J. O'Brien, A.K. Gilligan, L. Kuhlmann, Deep learning for automated epileptiform discharge detection from scalp EEG: a systematic review, *J. Neural. Eng.* 19 (5) (2022) 51002.
- [16] V. Singh, K. Veer, R. Sharma, S. Kumar, Comparative study of FIR and IIR filters for the removal of 50 Hz noise from the EEG signal, *Int. J. Biomed. Eng. Technol.* 22 (3) (2016) 250–257.
- [17] D. Bansal, R. Mahajan, Design and implementation of an efficient digital filter for preprocessing of EEG signals, in: *2019 6th International Conference on Computing for Sustainable Global Development (Indiacom)*, IEEE, 2019, March, pp. 862–868.
- [18] B. Burle, L. Spieser, C. Roger, L. Casini, T. Hasbroucq, F. Vidal, Spatial and temporal resolutions of EEG: is it really black and white? A scalp current density view, *Int. J. Psychophysiol.* 97 (3) (2015) 210–220.
- [19] M. Melinda, Yunidar Syahrial, Bahri Al, M. Irahmsyah, Finite impulse response filter for electroencephalogram waves detection, *Green Intell. Syst. Appl.* 2 (1) (Apr.2022) 7–19, <https://doi.org/10.53623/gisa.v2i1.65>. Find Other Styles.
- [20] S.F. Abbasi, M. Awais, X. Zhao, W. Chen, Automatic denoising and artifact removal from neonatal EEG, in: *BIBE 2019; the Third International Conference on Biological Information and Biomedical Engineering*, VDE, 2019, June, pp. 1–5.
- [21] A.M. Abdulghani, A.J. Casson, E. Rodriguez-Villegas, Compressive sensing scalp EEG signals: implementations and practical performance, *Med. Biol. Eng. Comput.* 50 (2012) 1137–1145.
- [22] L.P. Wijesinghe, D.S. Wickramasuriya, Ajith A. Pasqual, A generalized preprocessing and feature extraction platform for scalp EEG signals on FPGA, *IEEE Confer. Biomed. Eng. Sci. (IECBES)* (2014) 137–142, 2014.
- [23] Furio Bartoli, Sergio Cerutti, An optimal linear filter for the reduction of noise superimposed to the EEG signal, *J. Biomed. Eng.* 5 (4) (1983) 274–280.
- [24] Sarath Chandra Machavarapu, Manoj Kumar Mukul, Dhiraj Kumar, EEG classification based on variance, in: *2014 International Conference on Green Computing Communication and Electrical Engineering*, 2014.
- [25] J.G. Proakis, Manolakis, Pearson Education, *Digital Signal Processing Principles, Algorithms, and Applications*, 2007.
- [26] P. Podder, T. Zaman Khan, M. Haque Khan, M. Mukhtar Rahman, Comparative performance analysis of hamming, hanning and blackman window, *Int. J. Comput. Appl.* 96 (18) (2014) 1–7.
- [27] N.E. Diana, U. Kalsum, A. Sabiq, A. Jatmiko, P. Mursanto, Comparing windowing methods on finite impulse response (FIR) filter algorithm in electroencephalography (EEG) data processing, *J. Theor. Appl. Inf. Technol.* 88 (3) (2016) 558–567, 2016.
- [28] K. Majumdar, Human scalp EEG processing: various soft computing approaches, *Appl. Soft Comput.* 11 (8) (2011) 4433–4447.
- [29] R. Mehboob, S.A. Khan, R. Qamar, FIR filter design methodology for hardware optimized implementation, *IEEE Trans. Consum. Electron.* 55 (3) (2009) 1669–1673, <https://doi.org/10.1109/tce.2009.5278041>. Available:.
- [30] P. Das, S.K. Naskar, P. Narayan, Sankar, An approach to enhance performance of Kaiser window based filter, in: *2016 Second IEEE International Conference on Research in Computational Intelligence and Communication Networks (ICRCICN) (25th-26th Sep.2016)*, 2016, pp. 256–261, <https://doi.org/10.1109/ICRCICN.2016.7813666>. Available:.
- [31] M.A. Ahmad, N.A. Khan, W. Majeed, Computer-assisted analysis system of electroencephalogram for diagnosing epilepsy, in: *Proceedings of the 22nd International Conference on Pattern Recognition (ICPR)*, IEEE, 2014, pp. 3386–3391, <https://doi.org/10.1109/ICPR.2014.579>.
- [32] H.U. Amin, A.S. Malik, R.F. Ahmad, N. Badruddin, N. Kamel, M. Hussain, W.-T. Chooi, Feature extraction and classification for EEG signals using wavelet transform and machine learning techniques, *Australas. Phys. Eng. Sci. Med.* 38 (1) (2015) 139–149, <https://doi.org/10.1007/s13246-015-0333-x>.
- [33] M. Mursalin, Y. Zhang, Y. Chen, N.V. Chawla, Automated epileptic seizure detection using improved correlation-based feature selection with random forest classifier, *Neurocomputing* 241 (2017) 204–214, <https://doi.org/10.1016/j.neucom.2017.02.030>.
- [34] Y. Zhang, Y. Zhang, J. Wang, X. Zheng, Comparison of classification methods on EEG signals based on wavelet packet decomposition, *Neural Comput. Appl.* 26 (5) (2014) 1217–1225, <https://doi.org/10.1007/s00521-014-1776-y>.
- [35] R.R. Sharma, P. Varshney, R.B. Pachori, S.K. Vishvakarma, Automated system for epileptic EEG detection using iterative filtering, *IEEE Sensors Lett.* 2 (4) (2018) 1–4, <https://doi.org/10.1109/LENS.2018.2864738>.
- [36] N.D. Truong, L. Kuhlmann, M.R. Bonyadi, J. Yang, A. Faulks, O. Kavehei, Supervised learning in automatic channel selection for epileptic seizure detection, *Expert Syst. Appl.* 67 (2017) 340–355, <https://doi.org/10.1016/j.eswa.2016.09.046>.
- [37] M.H. Aslam, S.M. Usman, S. Khalid, A. Anwar, R. Alroobaea, S. Hussain, A. Yasin, Classification of EEG signals for prediction of epileptic seizures, *Appl. Sci.* 12 (14) (2022) 7251.
- [38] X. Wang, G. Gong, N. Li, S. Qiu, Detection analysis of epileptic EEG using a novel random forest model combined with grid search optimization, *Front. Hum. Neurosci.* 13 (2019) 12, <https://doi.org/10.3389/fnhum.2019.00012>.
- [39] R.S. Selvakumari, M. Mahalakshmi, P. Prashalee, Patient-specific seizure detection method using hybrid classifier with optimized electrodes, *J. Med. Syst.* 43 (5) (2019) 121, <https://doi.org/10.1007/s10916-019-1268-2>.
- [40] A.P. Sharma, K.A. Rao, L. Somappa, Hardware Optimization and Implementation of a 16-Channel Neural Tree Classifier for on-chip Closed-Loop Neuromodulation, *IEEE Transactions on Biomedical Circuits and Systems*, 2024.
- [41] G. Bouallegue, R. Djemal, K. Belwafi, Artificial EEG signal generated by a network of neurons with one and two dendrites, *Results Phys.* 20 (2021) 103699.
- [42] K. Belwafi, O. Romain, S. Gannouni, F. Ghaffari, R. Djemal, B. Ouni, An embedded implementation based on adaptive filter bank for brain-computer interface systems, *J. Neurosci. Methods* 305 (2018) 1–16.
- [43] K. Belwafi, S. Gannouni, H. Aboalsamh, H. Mathkour, A. Belghith, A dynamic and self-adaptive classification algorithm for motor imagery EEG signals, *J. Neurosci. Methods* 327 (2019) 108346.
- [44] S. Gannouni, K. Belwafi, H. Aboalsamh, Z. AlSamhan, B. Alebdi, Y. Almassad, H. Alobaedallah, EEG-based BCI system to detect fingers movements, *Brain Sci.* 10 (12) (2020) 965.
- [45] I. Ahmad, M. Zhu, Z. Liu, M. Shabaz, I. Ullah, M.C.F. Tong, et al., Multi-feature fusion-based convolutional neural networks for EEG epileptic seizure prediction in consumer internet of things, *IEEE Trans. Consum. Electron.* 70 (3) (2024) 5631–5643.
- [46] I. Ahmad, Z. Liu, L. Li, I. Ullah, S.T. Aboyeji, X. Wang, et al., Robust epileptic seizure detection based on biomedical signals using an advanced multi-view deep feature learning approach, *IEEE J. Biomed. Health Inform.* 28 (10) (2024) 5742–5754.
- [47] K. Belwafi, S. Gannouni, H. Aboalsamh, An effective zeros-time windowing strategy to detect sensorimotor rhythms related to motor imagery EEG signals, *IEEE Access* 8 (2020) 152669–152679.
- [48] A. Pant, A. Kumar, C. Verma, Z. Illés, Comparative exploration on EEG signal filtering using window control methods, *Results Control Optimiz.* 17 (2024) 100485.
- [49] A. Pant, A. Kumar, Exploring EEG signal processing for effective filtering and classification of epileptic seizures, *Discov. Electron.* 3 (1) (2026) 20.
- [50] A.B. Tufail, I. Ullah, A.U. Rehman, R.A. Khan, M.A. Khan, Y.K. Ma, et al., On disharmony in batch normalization and dropout methods for early categorization of Alzheimer's disease, *Sustainability* 14 (22) (2022) 14695.
- [51] K. Belwafi, F. Ghaffari, O. Romain, R. Djemal, An embedded implementation of home devices control system based on brain computer interface, in: *Proc. 26th Int. Conf. Microelectronics (ICM)*, IEEE, 2014, pp. 140–143.
- [52] K. Belwafi, R. Djemal, F. Ghaffari, O. Romain, B. Ouni, S. Gannouni, Online adaptive filters to classify left and right hand motor imagery, in: *Special Session on Smart Embedded Biomedical Devices for in Situ Physiological Signal Processing*, SciTePress, 2016, pp. 335–339.
- [53] K. Belwafi, A. Alsuwaidi, S. Mejri, R. Djemal, Brain-inspired signal processing for detecting stress during mental arithmetic tasks, *Brain Informat.* 12 (1) (2025) 34.
- [54] A. Pant, A. Kumar, Hanning FIR window filtering analysis for EEG signals, *Biomed. Anal.* 1 (2) (2024) 111–123.
- [55] I. Ahmad, S. Ayouni, F. Ahmad, H. Li, S.T. Aboyeji, H. Bilal, et al., An AI-driven interpretable multiview feature learning approach for EEG-based epileptic seizure detection, *IEEE J. Biomed. Health Inform.* (2025) 1–14.
- [56] A. Goel, A.K. Goel, A. Kumar, The role of artificial neural network and machine learning in utilizing spatial information, *Spatial Inform. Res.* 31 (3) (2023) 275–285.
- [57] A. Goel, A.K. Goel, A. Kumar, Performance analysis of multiple input single layer neural network hardware chip, *Multimed. Tool. Appl.* 82 (18) (2023) 28213–28234.
- [58] A. Kumar, Study and analysis of different segmentation methods for brain tumor MRI application, *Multimed. Tool. Appl.* 82 (5) (2023) 7117–7139.
- [59] A. Kumar, P. Sharma, M.K. Gupta, R. Kumar, Machine learning-based resource utilization and pre-estimation for network on chip (NoC) communication, *Wirel. Pers. Commun.* 102 (2018) 2211–2231.
- [60] S. Gupta, A. Kumar, Study on early accurate diagnosis and treatment of COVID-19 with smartphone tracking using bionics, *Secur. Privac.* 6 (5) (2023) e303.
- [61] S. Gupta, S.A. HariPrasad, A. Kumar, Bionic-based emergency contact for treatment of epilepsy, in: *Proceedings of the 2018 10th International Conference on Bioinformatics and Biomedical Technology*, 2018, May, pp. 46–50.